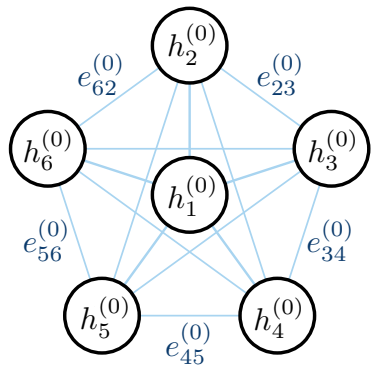


INPUT GRAPH



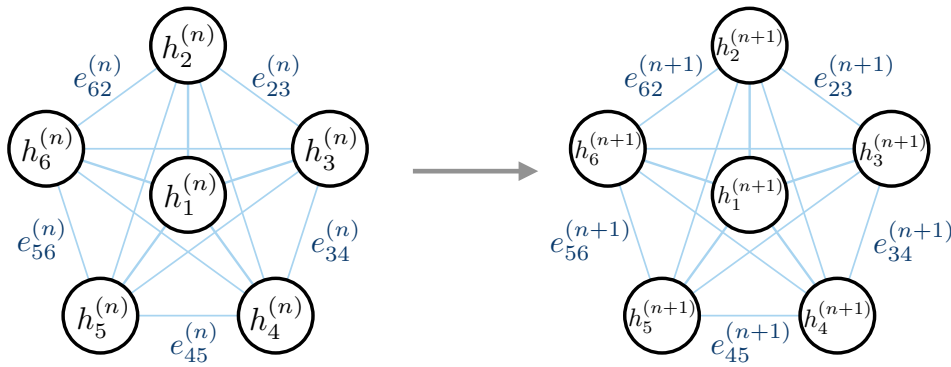
$$h_a^{(0)} = \hat{p}_a$$

initial node features

$$e_{ab}^{(0)} = \frac{\hat{\rho}_{ab} - \text{HD}_{ab}}{\sigma_{ab}}$$

initial edge features

CONVOLUTIONAL LAYERS



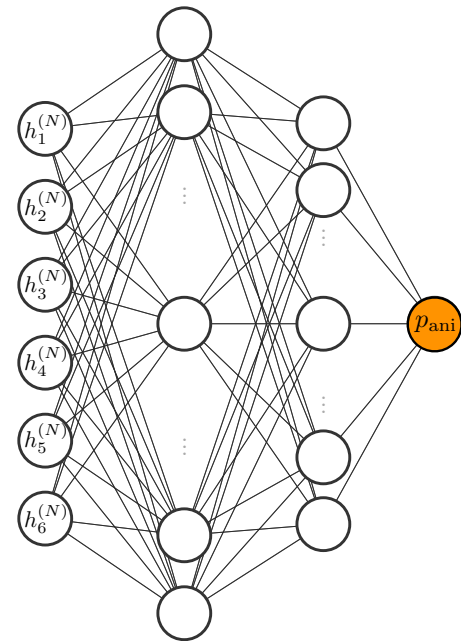
$$h_a^{(n+1)} = \text{MLP} \left[(1 + \varepsilon) h_a^{(n)} + \sum_{b \in \mathcal{N}(a)} \text{ReLU} \left(h_b^{(n)} + e_{ab} \right) \right]$$

node features are updated using a GINE convolution

$$e_{ab}^{(n+1)} = \text{MLP} \left(h_a^{(n+1)}, h_b^{(n+1)}, e_{ab}^{(n)} \right)$$

edge features are updated with a two-layer MLP

READOUT



final node features
are passed to MLP readout